



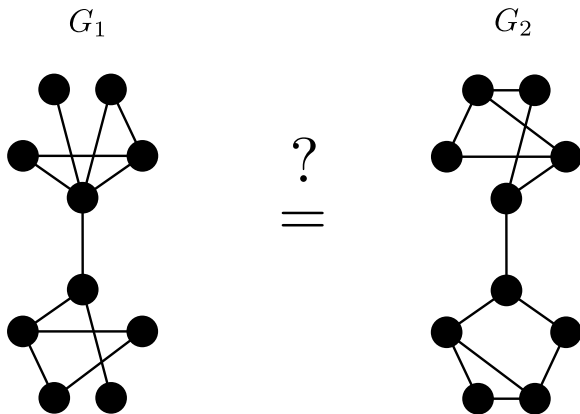
Network mutual information measures for graph similarity

Helcio Felippe

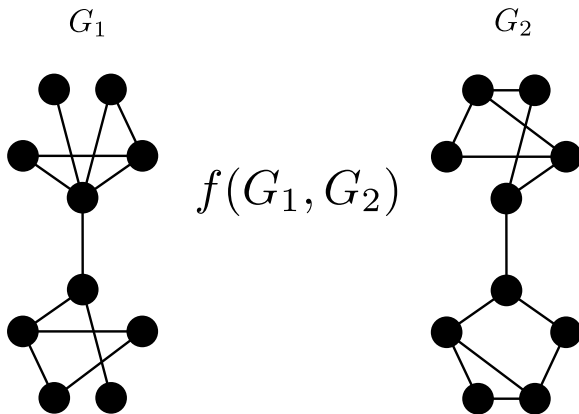
felippe_helcio@phd.ceu.edu

July 17, 2025

Graph similarity

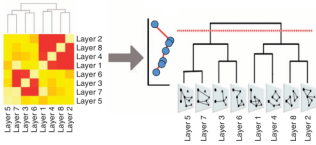


Graph similarity

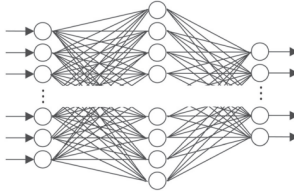


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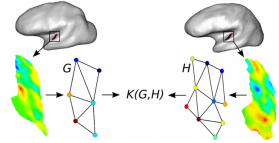
Applications to clustering, network regression, network classification.



M. De Domenico et al., *Nat. Commun.* (2015).



N. Attar et al., *Chaos* (2017).



S. Takerkart et al., *PLoS One* (2014).

Graph similarity

Applications to clustering, network regression, network classification.

	graph distance	label		
1	Jaccard [29]	JAC		
2	Hamming [30]	HAM		
3	Hamming–Ipsen–Mikhailov [37]	HIM	resistance perturbation [8]	REP
4	Frobenius [28]	FRO	NetLSD [3]	LSD
5	polynomial dissimilarity [5]	PDD	Lap. spectrum; Gauss. kernel, JSD [27]	LGJ
6	degree JSD [34]	DJS	Lap. spectrum; Loren. kernel, Euc. [27]	LLE
7	portrait divergence [4]	POR	Ipsen–Mikhailov [43]	IPM
8	quantum spectral JSD [40]	QJS	non-backtracking eigenvalue [7]	NBD
9	communicability sequence [41]	CSE	distributional non-backtracking [38]	DNB
10	graph diffusion distance [42]	GDD	D-measure distance [9]	DMD
			DeltaCon [2]	DCN
			NetSimile [1]	NES

H. Hartle et al., *Proc. R. Soc. A* (2020).

Drawbacks such as ad hoc methods, computationally intensive, and lack of multiscale similarity.

Network mutual information

Let G_1 and G_2 be graphs on the same set of N labeled nodes.

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$$\text{MI}(G_1, G_2) = \underbrace{\text{H}(G_2)}_{\text{information to transmit } G_2} - \underbrace{\text{H}(G_2|G_1)}_{\text{information to transmit } G_2 \text{ given } G_1} .$$

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The form of $\text{H}(\cdot)$ will depend on the particular encoding scheme used:

$$\text{H}(G_2) = \log \binom{N}{E_2}$$

$$\text{H}(G_2|G_1) = \log \binom{E_1}{E_{12}} \binom{N}{E_2} - E_{12}$$

$$\text{I}(G_1; G_2) = \text{H}_b(p_1) + \text{H}_b(p_2) - \text{H}_s(\mathbf{P}_{12})$$

$$\text{NMI}(G_1; G_2) = 2 \times \frac{\text{I}(G_1; G_2)}{\text{H}_b(p_1) + \text{H}_b(p_2)}$$

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The form of $\text{H}(\cdot)$ will depend on the particular encoding scheme used:

$$\begin{aligned}\text{H}(G_2) &= \log \binom{N}{E_2} \\ \text{H}(G_2|G_1) &= \log \binom{E_1}{E_{12}} \binom{\binom{N}{2} - E_1}{E_2 - E_{12}} \\ \text{I}(G_1; G_2) &= \text{H}_b(p_1) + \text{H}_b(p_2) - \text{H}_s(\mathbf{P}_{12})\end{aligned}$$

$$\begin{aligned}\text{H}_{\text{deg}}(G_2) &= \sum_{i=1}^N \log \binom{N}{k_2(i)} \\ \text{H}_{\text{deg}}(G_2|G_1) &= \sum_{i=1}^N \log \binom{k_1(i)}{k_{12}(i)} \binom{N - k_1(i)}{k_2(i) - k_{12}(i)}\end{aligned}$$

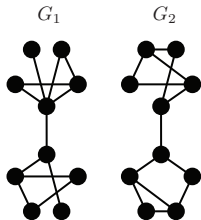
$$\begin{aligned}\text{H}(\tilde{G}_i^{(b)}) &= \log \left(\binom{\binom{B}{2} + B}{E_i} \right) \\ \text{H}(\tilde{G}_1^{(b)}, \tilde{G}_2^{(b)}) &= \log \left(\binom{\binom{B}{2} + B}{E_1 + E_2 - E_{12}^{(b)}} \right)\end{aligned}$$

$$\text{NMI}(G_1; G_2) = 2 \times \frac{\text{I}(G_1; G_2)}{\text{H}_b(p_1) + \text{H}_b(p_2)}$$

$$\text{DC-NMI}(G_1; G_2) = 2 \times \frac{\text{I}_{\text{deg}}(G_1; G_2)}{\frac{1}{N} \sum_i [\text{H}_b(p_1(i)) + \text{H}_b(p_2(i))]}$$

$$\text{MesoNMI}^{(b)}(G_1; G_2) = \frac{\text{I}_{\text{meso}}^{(b)}(G_1; G_2) - \text{I}_{\text{meso}}^{(b)}(E_{12}^{(b)} = 0)}{\frac{\text{H}(\tilde{G}_1^{(b)}) + \text{H}(\tilde{G}_2^{(b)})}{2} - \text{I}_{\text{meso}}^{(b)}(E_{12}^{(b)} = 0)}$$

Information-theoretic graph similarity measures

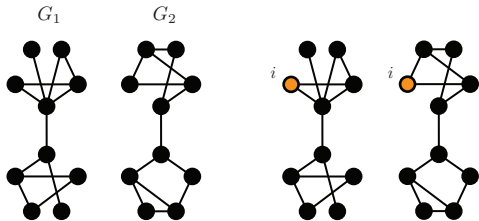


$$E_1 = 12, E_2 = 13$$

$$E_{12} = |G_1 \cap G_2| = 6$$

$$\text{NMI} = 0.06$$

Information-theoretic graph similarity measures



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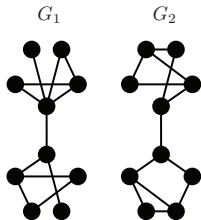
$$k_1(i) = 2, k_2(i) = 2$$

$$k_{12}(i) = |\partial_i^{(1)} \cap \partial_i^{(2)}| = 1$$

$$\text{NMI} = 0.06$$

$$\text{DC-NMI} = 0.18$$

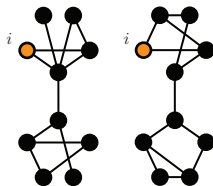
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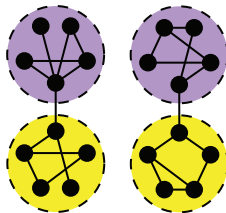
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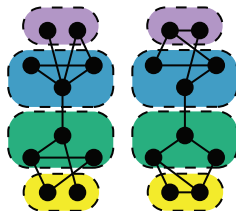
$$k_{12}(i) = |\partial_i^{(1)} \cap \partial_i^{(2)}| = 1$$

$$\text{DC-NMI} = 0.18$$



$$B = 2, E_{12}^{(b)} = |\tilde{G}_1^{(b)} \cap \tilde{G}_2^{(b)}| = 12$$

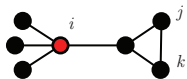
$$\text{MesoNMI} = 0.94$$



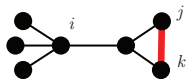
$$B = 4, E_{12}^{(b)} = 11$$

$$\text{MesoNMI} = 0.84$$

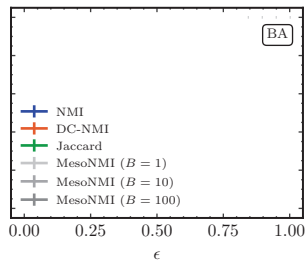
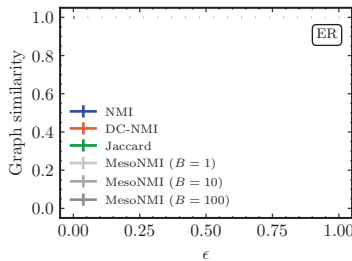
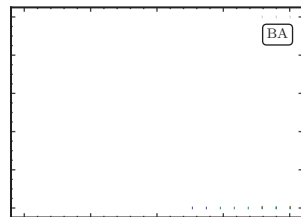
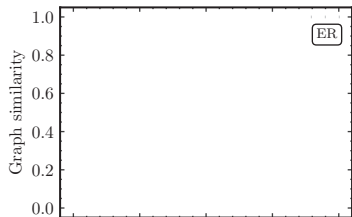
Test on synthetic graphs: ER and BA under attack



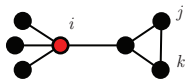
Node attack



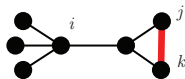
Edge attack



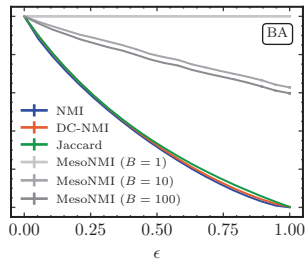
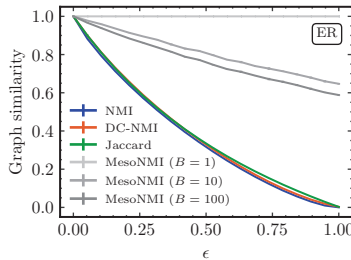
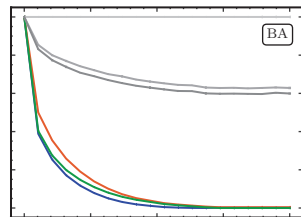
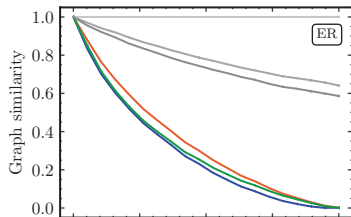
Test on synthetic graphs: ER and BA under attack



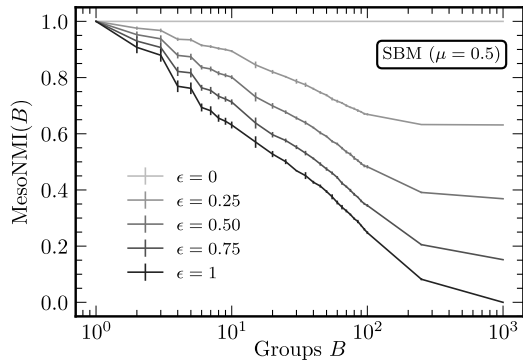
Node attack



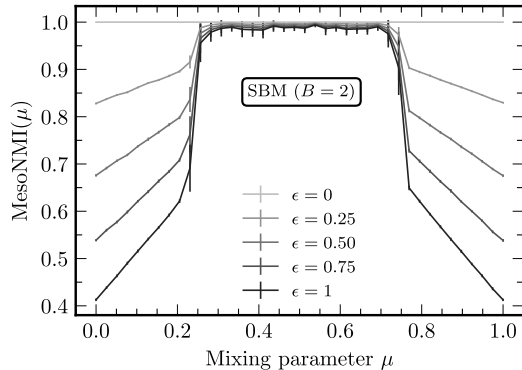
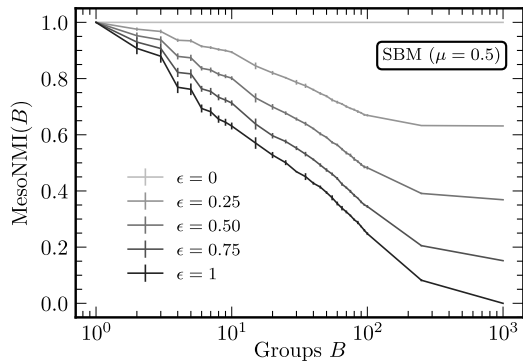
Edge attack



Mesoscale similarity for the SBM

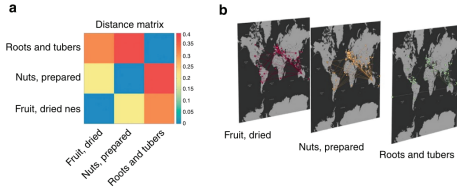


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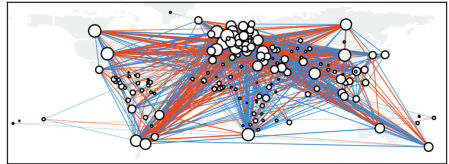


Case study: FAO trade network

Worldwide food import-export trade network of the Food and Agriculture Organization (FAO) of the United Nations.



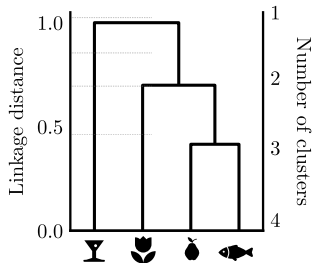
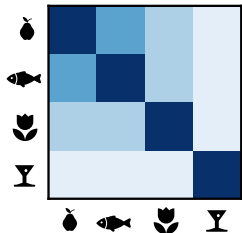
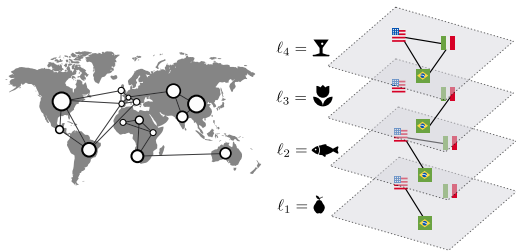
M. De Domenico et al., *Nat. Commun.* (2015).



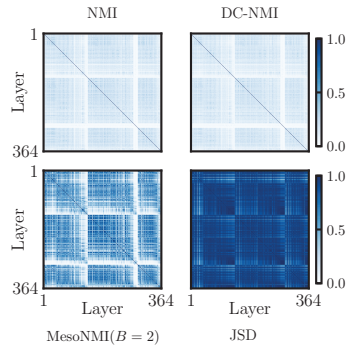
A. Kirkley et al., *Commun. Phys.* (2023).

Case study: FAO trade network

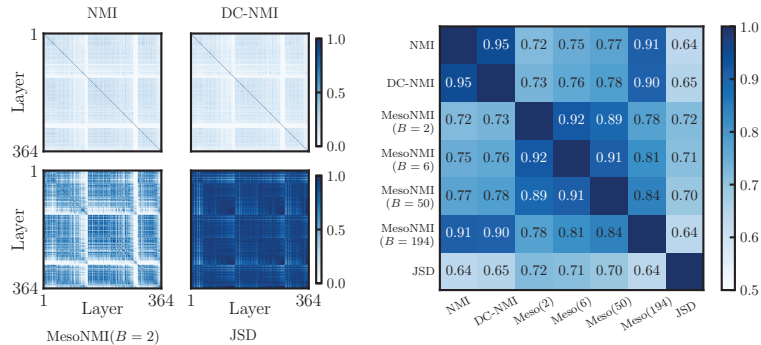
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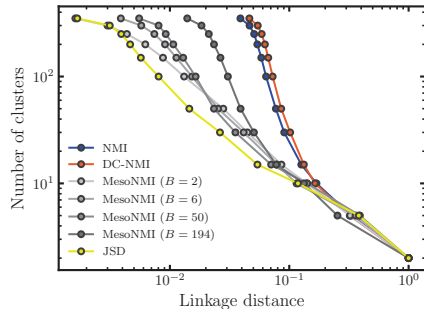
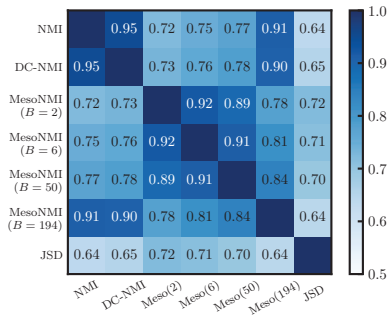
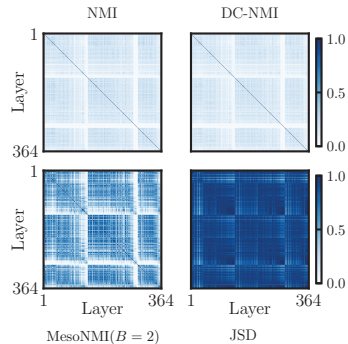
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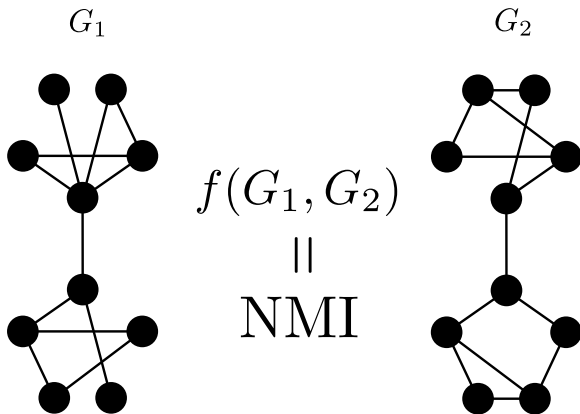
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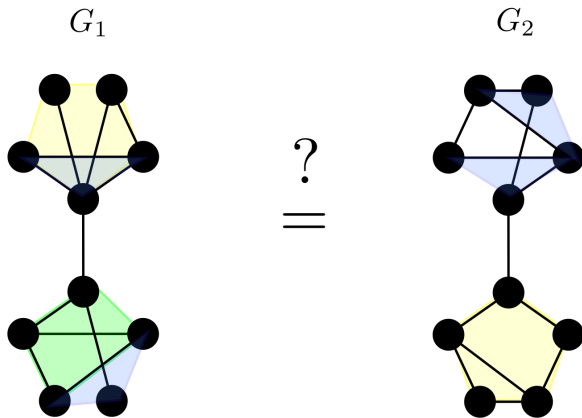
Graph similarity



network-MI/



Hypergraph similarity



Hypergraph similarity

The NMI is a special case for layer $\ell = 2$ of

$$H(G^{(\ell)}) = \log \binom{\binom{N}{\ell}}{E^{(\ell)}} .$$

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Analogously for layers ℓ and k ,

$$H(G_2^{(\ell)} | G_1^{(k)}) = \log \binom{E_1^{(k \rightarrow \ell)}}{E_{12}^{(k \rightarrow \ell, \ell)}} \binom{\binom{N}{\ell} - E_1^{(k \rightarrow \ell)}}{E_2^{(\ell)} - E_{12}^{(k \rightarrow \ell, \ell)}} .$$

Hypergraph similarity

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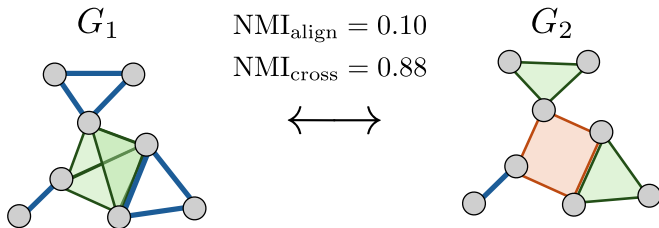
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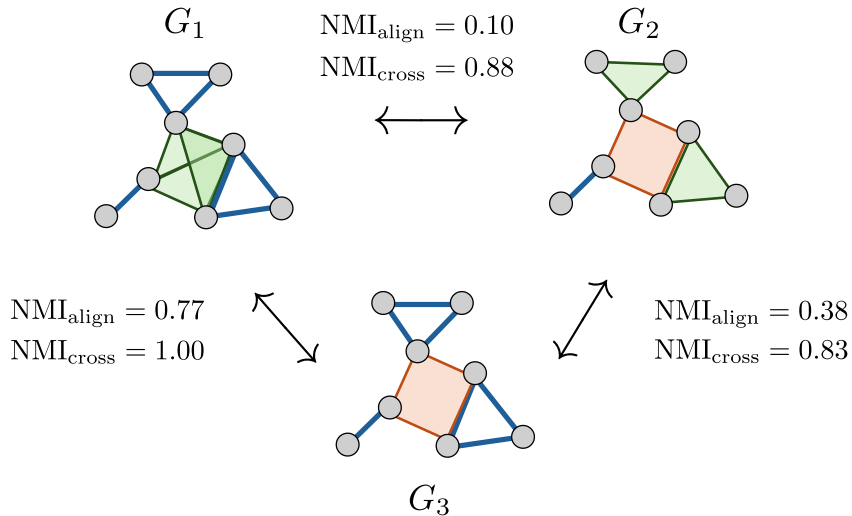
The NMI of hypergraphs G_1 and G_2 is thus

$$\text{NMI}(G_1, G_2) = \max \left\{ \frac{H(G_1) - H(G_1 | G_2)}{H(G_1)}, \frac{H(G_2) - H(G_2 | G_1)}{H(G_2)} \right\}.$$

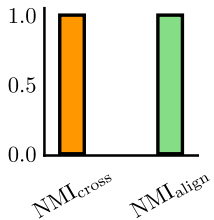
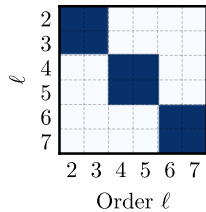
Hierarchy of hypergraph similarity measures



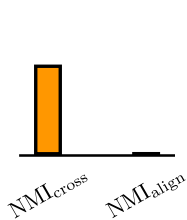
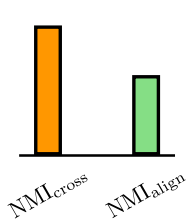
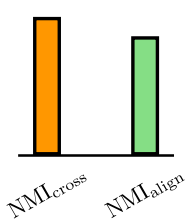
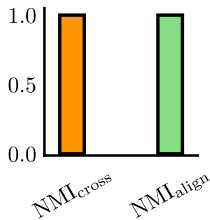
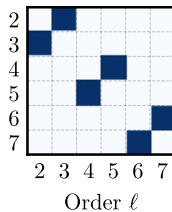
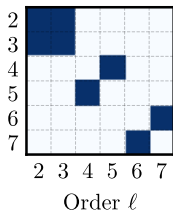
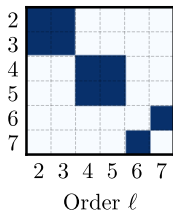
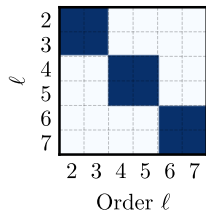
Hierarchy of hypergraph similarity measures



Intra- vs. inter-order similarity



Intra- vs. inter-order similarity



Final words

Efficient, multiscale, principled measure of graph and hypergraph similarity.

Extension to weighted, temporal, multilayer hypergraphs.



communications physics

Network mutual information measures for graph similarity

[Helcio Felippe](#), [Federico Battiston](#) ✉ & [Alec Kirkley](#) ✉

[Communications Physics](#) 7, Article number: 335 (2024) | [Cite this article](#)



Alec Kirkley

akirkley@hku.hk



Federico Battiston

battistonf@ceu.edu



Department of
Network and
Data Science

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THE UNIVERSITY OF HONG KONG

Thank you!

Back up slides

